

Rules and Regulations for 3KG Lego Sumo Category

1. About Competition

The 3KG Lego Sumo competition is an exciting challenge where participants create autonomous robots designed to push their opponents out of the ring. Success in the competition requires more than just raw power; competitors must carefully design their robots to be strong, stable, and agile while developing smart strategies to outmaneuver their opponents. Following the official rules of engagement is essential, and the combination of engineering skills, tactical thinking, and precise execution often determines the winner.

2. Robot Specification

1. Materials and Equipment:

- Robots must be built entirely from LEGO components. Custom-made or 3D-printed parts are strictly prohibited.
- Any LEGO wheels or tracks are allowed, including those from EV3, Spike, or LEGO Technic sets.
- Only one active LEGO controller (brick) may be used during the match, such as NXT, EV3, Spike, or 51515. A second controller may be used for building or programming purposes, but not during competition.
- There is no limit to the number of motors or sensors used, as long as they come from LEGO robotics kits like NXT, EV3, Spike. LEGO Technic motors that are not part of these kits are not permitted.

2. Size and Weight:

- The robot must not exceed a 20 cm × 20 cm boundary.
- The robot must not exceed a maximum weight of 3kg.
- No height limit.

3. Power and Signal Communication :

- The robot must work autonomously by power source.
- Wi-Fi and Bluetooth must be turned off during the match. The program must be uploaded before the match.

3. Match Regulations

1. Start:

- In total there will be 3 rounds per match. Both teams proceed to the ring to position their robots on the competition surface in accordance with the match setup procedure.

2. Match Duration:

- Each round will last a maximum of 3 minutes.
- If the robots fail to win within 3 minutes, then this round will be played again.

3. Victory Conditions:

- The robot wins if it pushes another robot away from the dohyo.
- If half of the robot goes beyond the white part of the dohyo, then this robot will lose and the point will be counted to the other robot.
- If both robots go beyond the white part of the dohyo the round will be counted as a draw and will be played again.

4. Match Procedure:

- The round will begin with a signal from the referee.
- Referees may stop the round if there are some problems.
- When the round begins, the robot shall observe a 5-second countdown after activation before initiating any movement.
- Connecting to the robot via Bluetooth or Wi-Fi is prohibited during matches. If anyone connects to the robot via Bluetooth or Wi-Fi, the team will be disqualified.
- The robot is permitted to deploy or unfold components after movement begins; however, it must remain a single, unified structure at all times and is not allowed to detach into separate parts.
- The robots must clearly display an identification number assigned by the organizers on the exterior casing in a clearly visible position.

5. Repairs and Maintenance

- Breaks between rounds are not permitted unless an unexpected or exceptional circumstance occurs.
- If a robot experiences a failure during a match, the referee may grant a 2-minute repair period. All repairs must be carried out under the supervision of an assistant referee to ensure that only approved components are used and no unauthorized modifications are made.
- If the robot is not repaired within 2 minutes, it will lose and the point will go to its opponent.

- A competitor may request a stoppage of the match in the event of personal injury or significant robot damage that prevents the match from continuing. A break will be granted if the referee determines it is necessary and justified.

4. Sportsmanship and Fair Play

1. Sportsmanship:

- Participants are expected to demonstrate fairness, respect, and good sportsmanship throughout the competition.
- Unsportsmanlike conduct may lead to disqualification. The Referee reserves the right to disqualify any supervisor, participant, or robot found to be acting unethically or in violation of sportsmanship standards. All decisions made by the Referee are final and not subject to appeal.

2. Inspections:

- Before the competition, the robots will be inspected to ensure they comply with rules
- Referees may ask you about the robot to ensure that the student built the robot by itself. Referees may ask you before or after the competition.

3. Participant behaviour:

- Participants are not allowed to touch the robot during the match.
- Having communication between participants and the referees is strictly prohibited
- During the game, 1 participant and 1 assistant must be present during the game. Other participants cannot be with them.

4. Maintenance:

- Participants may repair the robot before the match starts.
- Mentor and Assistant can help you and even give you advice during the game.
- If there's a problem with the robot during the match, the referee will give you 2 minutes to fix it. A mentor or assistant can help you.

5. Authority of the Judges:

- The Referee has the authority to supervise all matches and ensure that the competition rules are followed correctly by all participants.
- The Referee has the power to make final decisions regarding victories, disqualifications, rule violations, and match results during the competition.
- The Referee is responsible for maintaining fairness, safety, and discipline throughout the event and may stop a match if necessary.
- The Referee may provide warnings or penalties to participants who do not follow the official regulations and competition procedures.

5. Safety Rules

1. Safety of Robot:

- The robot must not have any parts that could harm the participant or robot.
- The referee will check your robot for safety, and if it's accepted, you'll be able to participate. If not, you'll have time to fix it.
- Participants are forbidden from using any device that can throw or discharge liquids, dust, gas, or other substances at an opponent.

2. Approval:

- All robots will be checked for safety to ensure compliance with the rules.
- If any safety issues are found with the robot, participants will have time to fix it or it will be disqualified.

6. Dohyo Specification

1. The ring consists of a circular platform with a diameter of 77 cm(770 mm).
2. The surface is black with a 2.5(25mm) cm white boundary line.
3. The robot **must not** damage the dohyo.

7. Disqualification

1. The robot will be disqualified if the robot does not comply with the rules.
2. The robot will be disqualified if it damages the robot or participant.
3. If a participant shouts or goes against the referee.
4. If a participant will disrespect other teams.
5. If the robot won't be fixed during the match.
6. The robot must not be physically modified or have its code changed after inspection.
7. If the participant won't come after the call-up.
8. If you are called to play and you don't come in 5 minutes the robot will lose the match